

2-Mark Questions & Answers -- MPCA Module 6: Pentium 4 & ARM Processor

Q1. What is NetBurst microarchitecture in Pentium 4

The NetBurst Microarchitecture is Intel's 7th-gen x86 design featuring a 20-stage hyper-pipeline, Execution Trace Cache, and Rapid Execution Engine to achieve high clock speeds (up to 3.8 GHz).

Q2. What is Execution Trace Cache in Pentium 4

The Execution Trace Cache replaces the traditional L1 instruction cache by storing up to 12,000 pre-decoded micro-operations (uops) in their predicted execution sequence, bypassing the instruction decoder.

Q3. Define Hyper-Threading Technology (HT).

Hyper-Threading is Simultaneous Multithreading (SMT) that allows a single physical processor core to function as two logical processors by duplicating architectural registers while sharing execution units.

Q4. What is the Inline Barrel Shifter in ARM processors

The Inline Barrel Shifter is an on-chip hardware unit in ARM that can shift or rotate one operand by arbitrary bit positions within the same single clock cycle as arithmetic/logical operations.